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AI-Driven Real-Time Early Warning System for Sand Blockage Mitigation in Hydraulic Fracturing Operations

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Abstract

Sand blockage, or screen-out, remains one of the most disruptive and costly challenges in hydraulic fracturing, particularly in deep, geologically complex shale formations. Traditional mitigation strategies—relying on human monitoring of surface pressure data—are often reactive and insufficient for preventing treatment interruptions. This paper introduces a novel AI-driven real-time sand blockage early warning system, integrated into the Testing and Fracturing Operation Center (TFOC) platform, which enables proactive intervention during fracturing operations. The system continuously ingests real-time pressure and pumping data, analyzes pressure trends using slope-based algorithms, and forecasts short-term blockage risks up to 120 seconds in advance. Upon detecting potential sand-out scenarios, the system automatically issues alerts and provides operators with actionable recommendations such as reducing slurry rate or halting proppant injection. Field applications across horizontal shale wells in China—including detailed analysis of five critical time points and four additional case studies—demonstrate that the system enhances operational safety, reduces non-productive time (NPT), and improves stimulation effectiveness. Results show that the system not only enables timely adjustments to avoid severe blockages but also supports complex fracture network generation, making it a transformative tool for intelligent fracturing operations in unconventional reservoirs.

Introduction

Hydraulic fracturing remains the cornerstone of unconventional oil and gas development, enabling the economic extraction of hydrocarbons from low-permeability formations such as shale, tight sandstone, and coalbeds. By injecting high-pressure fluids and proppants into the reservoir, artificial fractures are generated, significantly enhancing flow conductivity and increasing recovery factors (Warpinski et al., 2009). The evolution of hydraulic fracturing technology—from single-stage vertical applications to multi-stage, extended-reach horizontal wells—has revolutionized the energy industry. Despite these advancements, the operational complexity of stimulation treatments has grown, especially in ultra-deep and structurally heterogeneous plays.

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Among the most persistent and costly operational challenges in hydraulic fracturing is sand blockage, or screen-out (Cleary et al., 1993; Dontsov and Peirce, 2014; Vo et al., 2022). This occurs when proppant-laden fluid encounters elevated resistance in the fracture system, leading to a rapid rise in treating pressure and interruption of pumping operations. The onset of sand blockage can damage surface equipment, increase non-productive time (NPT), compromise fracture conductivity, and negatively impact well performance (Barree and Conway, 1995; Cipolla et al., 2011). In particular, formations with intricate natural fracture networks, such as those in the Sichuan Basin, exacerbate the unpredictability of slurry transport and fracture containment. The basin's geological profile—characterized by high stress anisotropy, deep burial (often exceeding 3,500 meters), and the presence of multiple bedding planes and faults—makes sand blockage events especially difficult to predict and mitigate (Yong et al., 2019; Wu et al., 2023, 2024; Zeng et al., 2021).

Traditional sand blockage mitigation relies on real-time monitoring of wellhead pressure and pumping parameters by field engineers. However, the human ability to interpret pressure trends and respond proactively is limited by data latency and the complexity of subsurface conditions. Most decisions are reactive, based on pressure spikes that have already exceeded safe thresholds. While some diagnostic tools such as downhole pressure gauges and microseismic imaging have been employed, they typically lack the responsiveness and predictive power necessary for real-time intervention. Thus, there is an urgent need for intelligent, automated systems that can analyze live operational data, predict abnormal behavior, and initiate early warnings.

This paper presents a novel AI-driven sand blockage early warning system that has been successfully implemented as a module within the Testing and Fracturing Operation Center (TFOC) platform—a real-time digital fracturing system. The TFOC platform integrates real-time fracture simulation, 3D geomechanical visualization, automated data ingestion, and AI calibration modules to optimize stimulation in complex reservoirs. The sand blockage early warning module complements these capabilities by continuously monitoring live pressure and pump rate data, computing pressure trends, and forecasting short-term escalation risks. Through predictive analytics, the system enables timely adjustments to pumping operations, significantly reducing the risk of screen-out and enhancing fracture stage efficiency. Field results from over 100 horizontal wells have demonstrated the tool's robustness, reliability, and value as a transformative technology in intelligent hydraulic fracturing.

Methodology

The sand blockage early warning system is a predictive analytics module integrated within the TFOC (Testing and Fracturing Operation Center) platform. TFOC is a real-time, cloud-native fracturing optimization system developed to support data-driven stimulation operations across China's shale gas fields, as shown in Figure 1. The platform's layered architecture consists of a data ingestion layer (connected to regional cloud infrastructure), a computational layer with embedded simulation engines, and an application layer that includes AI modules for fracture geometry calibration, treatment optimization, and risk detection. The sand blockage algorithm functions as part of this application layer and leverages live field data streamed directly from pressure sensors and slurry pumps.

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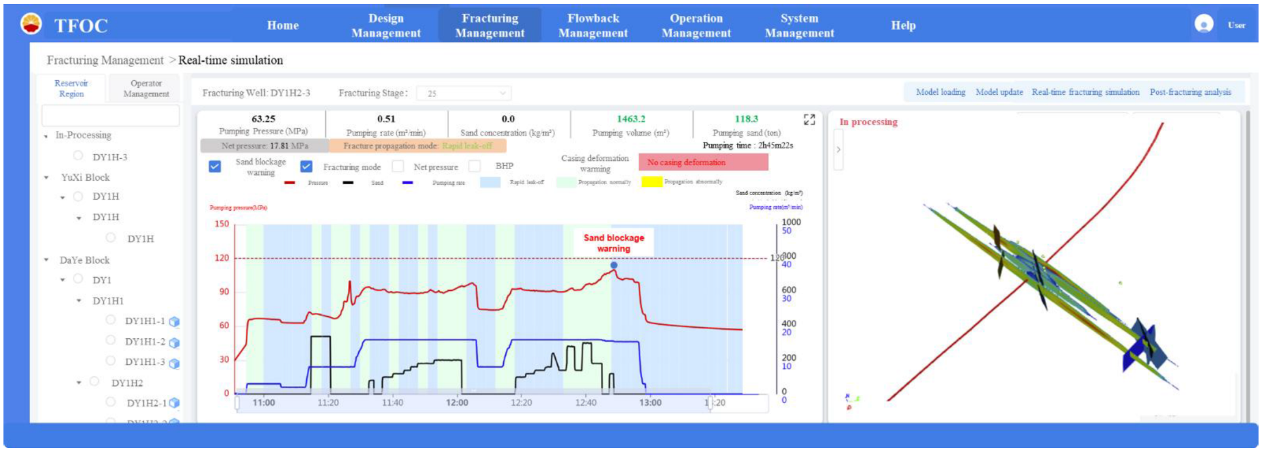

Figure 1—A real-time, cloud-native testing and fracturing operation center (TFOC) platform.

The core logic of the early warning system is based on real-time slope analysis of wellhead casing pressure, as shown in Figure 2. The algorithm initiates analysis only when two operational thresholds are met: (1) total sand injection exceeds 5 metric tons, and (2) pump rate is above 5 m³/min, ensuring that only relevant data is analyzed. Once these conditions are satisfied, the system continuously evaluates the pressure behavior using the following steps:

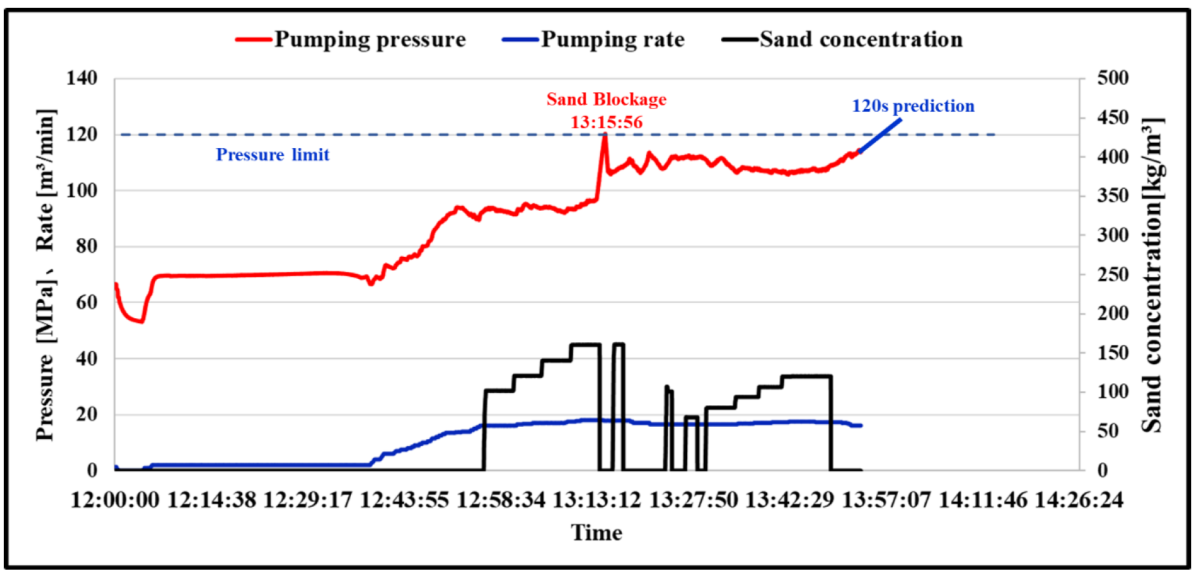

Figure 2—Sand blockage warning system based on real-time slope analysis of wellhead casing pressure.

Data Collection and Pre-Condition Checks

The system retrieves casing pressure at the current timestamp p_n and 10 seconds prior p_{n-1}. If the current pressure p_n already exceeds the predefined wellhead limit pressure p_{limit} (set at 120 MPa for deep shale gas formations and 70 MPa for shallow shale gas formations), the system immediately issues a blockage warning and logs the event.

Slope-Based Pressure Trend Calculation

If $p_n < p_{limit}$, the system calculates the rate of pressure change (slope) using:

$$s = \frac{p_n - p_{n-1}}{t_n - t_{n-1}} = \frac{p_n - p_{n-1}}{10 \text{ seconds}} \quad (1)$$

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A positive slope s indicates increasing pressure and potential flow resistance buildup, initiating forecasting procedures.

Short-Term Pressure Forecasting

The next 2 minutes are divided into twelve 10-second intervals (k is interval number). The casing pressure for each future point (p_{n+k}) is predicted using:

$$p_{n+k} = p_n + s \times (10 \times k) \quad k = 1, 2, \cdots, 12 \tag{2}$$

This linear extrapolation provides a forward-looking view of pressure evolution.

Early Warning Decision Logic

If any predicted value p_{n+k} equals or exceeds pressure limit p_{limit}, the system issues a "Sand Blockage" alert with the timestamp and predicted risk horizon. The warning is displayed on the TFOC user interface and logged into the stimulation summary report.

Alert Persistence and Logging

If multiple predicted values exceed the threshold, the earliest alert time is retained on-screen until the pressure drops below the limit. All alert events, including date and time, are compiled in the TFOC "Stimulation Summary" log, e.g., "4 alerts: 2024/06/21 12:35, 12:56, 13:05, 13:12".

Reinitialization and Continuous Monitoring

Once the risk passes, the system resets and resumes monitoring. The loop repeats throughout the entire stage to ensure uninterrupted risk detection.

Field Application

To evaluate the practical utility of the AI-driven real-time sand blockage early warning system, five key operational moments are highlighted below based on one field case study. These moments demonstrate how the system can be tightly integrated into field fracturing operations, providing timely alerts that enable on-site engineers to dynamically adjust pumping parameters—such as reducing slurry rate, pausing proppant injection, or modifying fluid properties—in order to prevent or localize sand blockage. This capability ensures each fracturing stage is executed efficiently, reducing risk and non-productive time.

Moment 1: Early Detection of Potential Sand Blockage

Following the initiation of proppant injection, a persistent and slight increase in wellhead treating pressure is observed, as shown in Figure 3. This subtle pressure rise indicates that sand blockage may be developing in the wellbore. The system's sand blockage early warning algorithm automatically forecasts the wellhead pressure trajectory over the next 120 seconds. The prediction shows the pressure approaching the critical sand-out threshold. Based on this early warning, the field team is prompted to make immediate operational adjustments—specifically, to halt proppant injection and modify the fracturing fluid properties—to prevent a full-scale sand-out event. The interface also integrates a real-time fracture propagation module with boundary element method (Wu and Olson, 2015), which aids engineers in analyzing the spatial dynamics contributing to the blockage risk.

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Figure 3—Sand blockage warning system at time 1 of field case 1.

Moment 2: Operational Response and Recovery

As illustrated in Figure 4, the early warning system triggers a visual alert indicating a high sand-out risk. In response, the field crew halts proppant injection and adjusts the fluid system accordingly. These actions quickly stabilize the wellhead pressure, bringing it back within the safe operating window. After maintaining a stable fluid-only injection for a short period, proppant injection is resumed. The wellhead pressure remains within a normal range, and no blockage occurs, validating the effectiveness of the early intervention.

Figure 4—Sand blockage warning system at time 2 of field case 1.

Moment 3: Renewed Risk Detected During Continued Proppant Injection

Figure 5 shows a subsequent stage in which continuous proppant injection leads to another minor rise in treating pressure. The AI system recalculates and predicts a potential sand-out event within the next 100 seconds. A timely early warning is issued, allowing the site crew to remain alert and prepared to modify the pumping schedule if needed.

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Figure 5—Sand blockage warning system at time 3 of field case 1.

Moment 4: Sand Blockage Confirmed Despite Intervention

Despite previous warnings and corrective measures—such as reducing the slurry rate and halting proppant injection—wellhead pressure continues to rise and remains elevated, as seen in Figure 6. This suggests that a downhole sand blockage has already occurred. The system correctly identifies the onset and severity of the blockage through continuous monitoring and trend recognition, providing critical information for real-time decision-making.

Figure 6—Sand blockage warning system at time 4 of field case 1.

Moment 5: Controlled Mitigation and Stage Completion

To ensure stage completion without escalating the blockage, the field team switches to a lower-rate fluid-only pumping strategy. Over time, this alleviates the blockage, and the fracturing operation proceeds to completion, as shown in Figure 7. However, the final proppant volume reaches only 131.8 tons, falling short of the originally designed 150 tons. This scenario underscores the practical value of the early warning system. By offering a 120-second foresight into blockage risk, the system enables proactive adjustments that control the severity of the event and ensure the stage is completed safely. Furthermore, real-time fracture propagation results show that a complex fracture network was successfully generated during this stage, with multiple natural fractures activated—demonstrating that despite the reduced proppant volume, the stimulation was effective.

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Figure 7—Sand blockage warning system at time 5 of field case 1.

Other Field Cases

To further demonstrate the practical value and robustness of the AI-driven real-time sand blockage early warning system, four representative field cases are presented below. These examples reflect different operational scenarios and showcase how the system guided field decisions to mitigate or respond to potential sand-out risks.

Case Study 2: Multiple Early Warnings with Successful Operational Adjustments (Figure 8). In this operation, the system issued five separate sand blockage early warnings throughout the stage. Upon receiving the first alert, the field crew responded by reducing the sand concentration, which successfully alleviated the early signs of blockage. For the remaining four warnings, the crew implemented a variety of adaptive strategies, including both reducing and increasing the slurry rate in response to evolving well conditions. These timely adjustments enabled the successful completion of the stage. However, the total proppant volume reached 110.27 tons, falling short of the originally planned 150 tons, demonstrating the trade-off between operational safety and sand placement.

Figure 8—Sand blockage warning system of field case 2.

Case Study 3: Late-Stage Blockage Despite Early Intervention (Figure 9). Initially, both fluid and proppant injection proceeded smoothly. However, as the stage progressed, the system issued warnings indicating rising blockage risk. In response, the field team implemented a series of preventive actions, including halting proppant injection and reducing the pump rate. Despite these efforts, wellhead pressure remained elevated and eventually spiked, confirming the occurrence of a downhole sand blockage. While

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the team intended to resume sand injection, the blockage prevented completion according to the original plan, highlighting the importance of early and aggressive mitigation in late-stage operations.

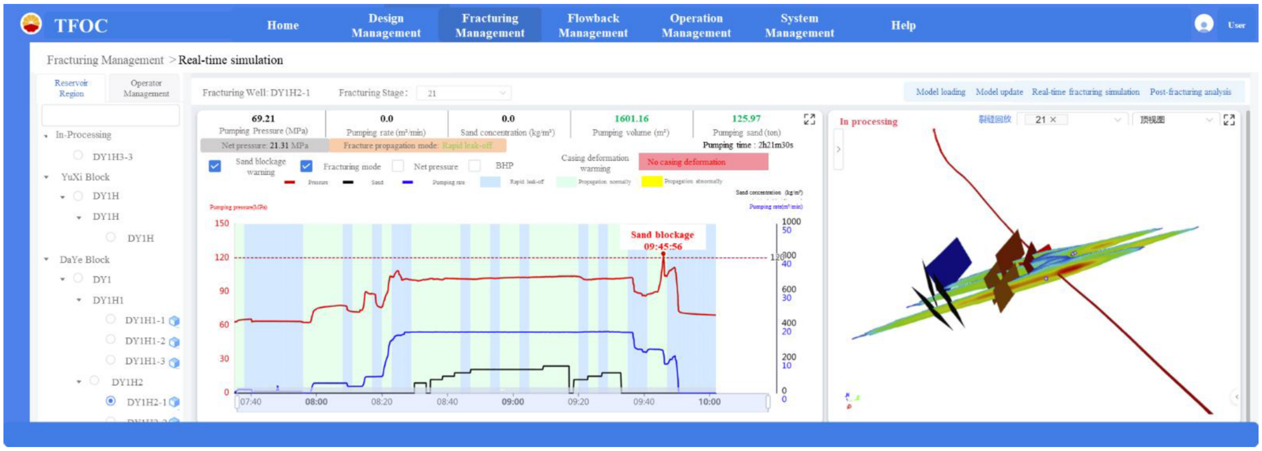

Figure 9—Sand blockage warning system of field case 3.

Case Study 4: Proactive Plan Adjustment Based on Risk Forecast (Figure 10). In this case, the operation proceeded with stable early-stage execution. However, the system forecasted three high-risk sand blockage events during the later stages of the treatment. Informed by these predictions, the field team preemptively modified the pumping schedule, including reducing the injection rate and lowering the proppant intensity. These proactive adjustments enabled safe completion of the stage, although the total proppant placed was 101.2 tons, below the designed 120 tons. This case underscores the value of predictive alerts in reshaping operational decisions before blockage occurs.

Figure 10—Sand blockage warning system of field case 4.

Case Study 5: Persistent Blockage Despite Mitigation Efforts (Figure 11). Late in the treatment, the system detected two sand blockage risk events as fracture pressure increased significantly. The site crew responded by lowering both the pump rate and sand concentration, but these adjustments were insufficient to fully relieve the blockage. While the stage was eventually completed, only 156.7 tons of proppant were injected, compared to the design target of 180 tons. This case highlights the system's role in diagnosing persistent blockage scenarios and guiding safe completion under compromised conditions.

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Figure 11—Sand blockage warning system of field case 5.

These diverse field examples reinforce the AI system's value in guiding real-time decision-making during hydraulic fracturing operations. By offering accurate short-term pressure trend forecasting and early blockage detection, the system empowers engineers to adapt dynamically, reduce the severity of sand-out events, and improve overall operational resilience.

Conclusions and Future Work

This study presents a field-validated, AI-driven early warning system for real-time detection and mitigation of sand blockage in hydraulic fracturing operations. Embedded within the cloud-native TFOC platform, the system continuously analyzes live casing pressure and pump rate data, forecasting critical pressure escalation events before they occur. By leveraging slope-based trend analysis and predictive extrapolation, the system provides engineers with up to 120 seconds of lead time to intervene—offering a significant advantage over traditional reactive monitoring approaches.

The five-stage operational walkthrough and four additional field cases illustrate a wide range of blockage risk scenarios and operational responses, from early detection and successful mitigation to persistent sand-out despite intervention. Across all applications, the system proved instrumental in guiding real-time decision-making, enhancing stage completion success rates, and preserving fracture complexity, even when proppant volumes were below design targets.

Ultimately, this AI-based early warning system elevates the intelligence and responsiveness of hydraulic fracturing workflows, particularly in ultra-deep and naturally fractured formations. Its predictive capability not only minimizes the operational and economic impacts of sand blockage but also fosters safer, more efficient, and more adaptable field practices—marking a key step forward in the digital transformation of stimulation operations.

While the current system focuses on real-time detection and short-term forecasting based on wellhead pressure trends, several enhancements are envisioned for the next stage of development: (1) Deep Learning-Based Risk Classification: By training deep learning models on historical stimulation data, the system will evolve to classify blockage types and suggest differentiated mitigation strategies automatically. (2) Closed-Loop Control Integration: The warning system will be extended to support semi-automated control, enabling real-time adjustments to pump schedules and blender rates through direct communication with surface equipment, under operator supervision. (3) Digital Twin Synchronization: A real-time digital twin of the fracture stimulation process will be developed to synchronize with AI alerts, providing a visual and analytical framework to assess both immediate blockage risks and long-term stimulation efficiency.

By combining AI, physics-based modeling, and cloud-based architecture, this real-time warning system offers a scalable and intelligent framework for managing one of the most challenging aspects of modern

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fracturing operations. Its continued evolution will further solidify the role of digital intelligence in enhancing stimulation reliability, reservoir access, and overall economic performance.

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